कुछ अभ्यासार्थ प्रश्नों के उत्तर

एकक 7

7.1 (i) 2,2,4-ट्राइमेथिलपेन्टेन –3-ऑल (ii) 5-एथिलहेप्टेन –2, 4-डाइऑल

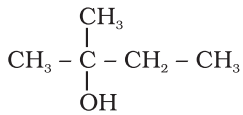
(iii) प्रोपेन –2,3-डाइऑल (iv) प्रोपेन –1,2,3,-ट्राइऑल

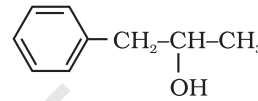
(v) 2- मेथिलफ़ीनॉल (vi) 4-मेथिलफ़ीनॉल

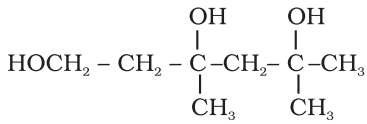
(vii) 2,5 – डाइमेथिलफ़ीनॉल (viii) 2,6-डाइमेथिलफ़ीनॉल

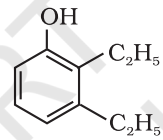
(ix) 1-मेथॉक्सी-2-मेथिलप्रोपेन (x) एथॉक्सीबेन्जीन

(xi) 1-फ़ीनॉक्सीहेप्टेन (xii) 2 –एथॉक्सीब्यूटेन

7.2 (i) 

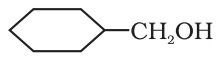
(ii) 

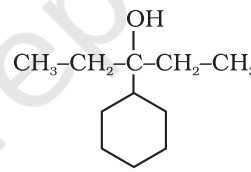
(iii) 

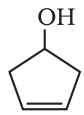
(iv) 

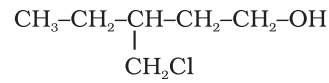
(v)

(vi) 

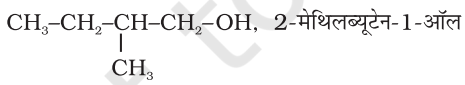
(vii) 

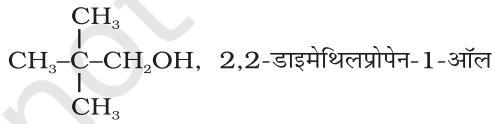
(viii) 

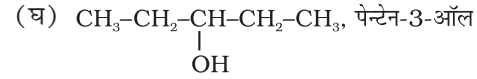
(ix) 

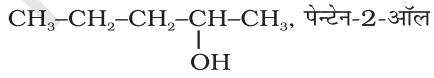
(x) 

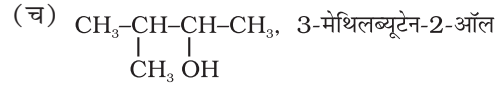
7.3 (क) $CH_3CH_2CH_2CH_2CH_2OH$, पेन्टेन-1-ऑल

(ख) 

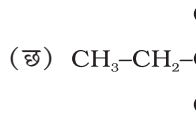
(ग) 

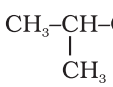
(घ) 

(ङ) 

(च) 

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(छ)  2-मेथिलब्यूटेन-2-ऑल

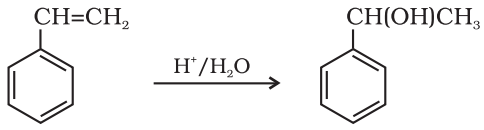
(ज)  3-मेथिलब्यूटेन-1-ऑल

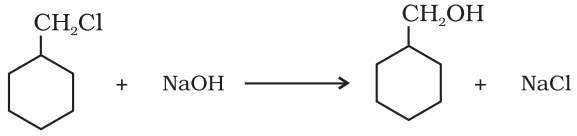
7.4 प्रोपेनॉल में हाइड्रोजन आबंधन

7.5 जल एवं ऐल्कोहॉल अणुओं के बीच आबंधन।

7.8 o-नाइट्रोफ़ीनॉल अंतराआण्विक हाइड्रोजन आबंधन के कारण भाप में वाष्पशील है।

7.12 **संकेत:** सल्फोनेशन के पश्चात नाभिकरागी प्रतिस्थापन करें।

7.13 (i) 

(ii) 

(iii) $CH_3(CH_2)_4Cl + NaOH \longrightarrow CH_3(CH_2)_4OH + NaCl$

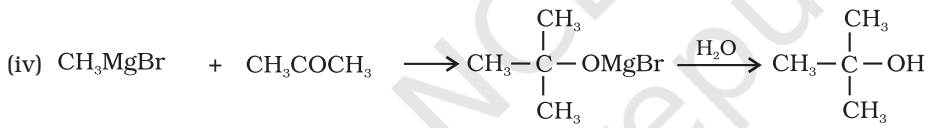
7.14 (i) सोडियम तथा (ii) सोडियम हाइड्रॉक्साइड के साथ अभिक्रिया

7.15 नाइट्रो समूह की इलेक्ट्रॉन खींचने की प्रवृत्ति और मेथॉक्सी समूह की इलेक्ट्रॉन विमोचक प्रवृत्ति के कारण

7.20 (i) प्रोपीन का जलयोजन

(ii) बेन्जिल क्लोराइड के –Cl का तनु NaOH के उपयोग द्वारा नाभिकरागी प्रतिस्थापन

(iii) $C_2H_5MgBr + HCHO \rightarrow C_2H_5CH_2OMgBr \xrightarrow{H_2O} C_2H_5CH_2OH$

(iv) 

7.23 (i) 1-एथॉक्सी-2-मेथिलप्रोपेन

(ii) 2-क्लोरो-1-मेथॉक्सीएथेन

(iii) 4-नाइट्रोऐनिसॉल

(iv) 1-मेथाक्सीप्रोपेन

(v) 1-एथाक्सी-4,4-डाइमेथिलसाइक्लोहेक्सेन

(vi) एथॉक्सीबेन्जीन

एकक 8

8.2 (i) 4-मेथिलपेन्टेनैल

(ii) 6-क्लोरो-4-एथिलहेक्सेन-3-ओन

(iii) ब्यूट-2-इनैल

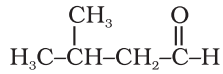
(iv) पेन्टेन-2,4-डाइओन

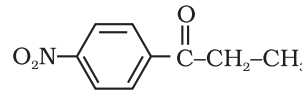
(v) 3,3,5-ट्राइमेथिलहेक्सेन-2-ओन

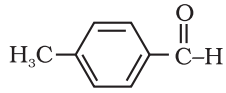
(vi) 3,3-डाइमेथिलब्यूटेनॉइक अम्ल

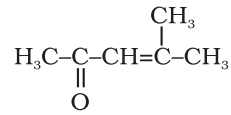
(vii) बेन्जीन –1,4-डाइकार्बेल्डिहाइड

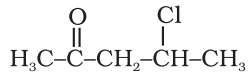
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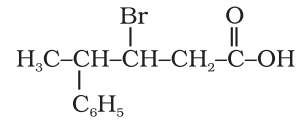
8.3 (i) 

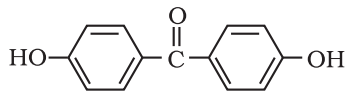
(ii) 

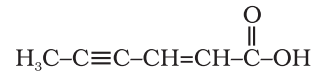
(iii) 

(iv) 

(v) 

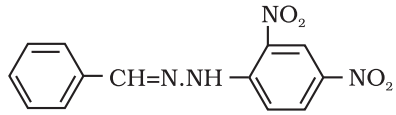
(vi) 

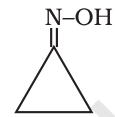
(vii) 

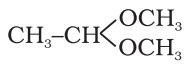
(viii) 

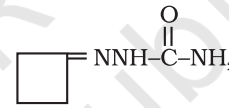
8.4 (i) हेप्टेन-2-ओन (ii) 4-ब्रोमो-2-मेथिलहेक्सेनैल (iii) हेप्टेनैल

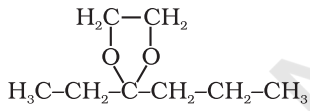
(iv) 3-फ़ेनिलप्रोप-2-ईनैल (v) साइक्लोपेन्टेनकार्बेल्डिहाइड (vi) डाइफ़ेनिलमेथेनोन

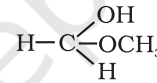
8.5 (i) 

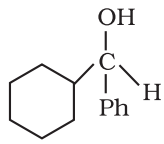
(ii) 

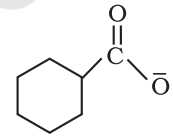
(iii) 

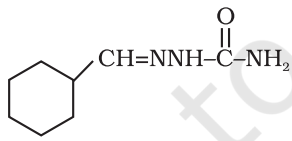
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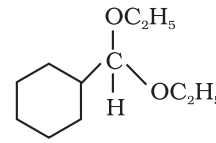
(v) 

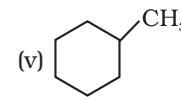
(vi) 

8.6 (i) 

(ii) 

(iii) 

(iv) 

(v) 

8.7 (ii), (v), (vi), (vii) – ऐल्डोल संघनन. (i), (iii), (ix) कैनिज़ारो अभिक्रिया (iv), (viii) कोई भी नहीं

8.10 2-एथिलबेन्जैल्डिहाइड (संरचना स्वयं लिखें)

8.11 (क) $CH_3CH_2CH_2COOCH_2CH_2CH_2CH_3$, ब्यूटिल ब्यूटेनोऐट

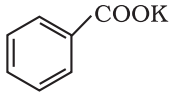
(ख) $CH_3CH_2CH_2$ COOH (ग) $CH_3CH_2CH_2CH_2OH$. समीकरण स्वयं लिखें

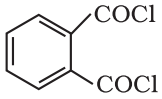
8.12 (i) डाइ-*तृतीयक*-ब्यूटिल कीटोन < मेथिल *तृतीयक*-ब्यूटिल कीटोन < ऐसीटोन < एसिटैल्डिहाइड

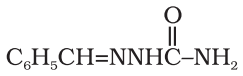
(ii) $(CH_3)_2CHCOOH$ < $CH_3CH_2CH_2COOH$ < $CH_3CH(Br)CH_2COOH$ < $CH_3CH_2CH(Br)COOH$

(iii) 4-मेथॉक्सीबेन्ज़ोइक अम्ल < बेन्ज़ोइक अम्ल < 4-नाइट्रोबेन्ज़ोइक अम्ल < 3,4-डाइनाइट्रोबेन्ज़ोइक अम्ल

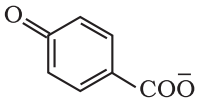
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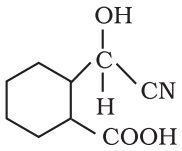
8.17 (i) 

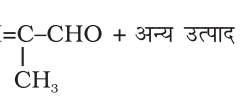
(ii) 

(iii) 

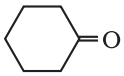
(iv) $C_6H_5COCl–AlCl_3$

(v) 

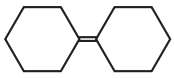
(vi) 

(vii) 

(viii) $CH_3CH(OH)CH_2COOC_2H_5$

(ix) 

(x) 1. BH_3; 2. $H_2O_2/\bar{O}H$; 3. PCC

(xi) 

8.19 यौगिक मेथिल कीटोन है और इसकी संरचना होगी– $CH_3COCH_2CH_2CH_3$

एकक 9

9.1 (i) 1-मेथिलएथिलऐमीन

(ii) प्रोपेन-1-ऐमीन

(iii) N-मेथिल-2-मेथिलएथिलऐमीन

(iv) 2-मेथिलप्रोपेन-2-ऐमीन

(v) N-मेथिलबेन्जेनेमीन या N-मेथिलऐनिलीन

(vi) N-एथिल-N-मेथिलएथेनेमीन

(vii) 3-ब्रोमोऐनिलीन या 3-ब्रोमोबेन्जेनेमीन

9.4 (i) $C_6H_5NH_2 < C_6H_5NHCH_3 < C_2H_5NH_2 < (C_2H_5)_2NH$

(ii) $C_6H_5NH_2 < C_6H_5N(CH_3)_2 < CH_3NH_2 < (C_2H_5)_2NH$

(iii) (a) *p* -नाइट्रोऐनिलीन < ऐनिलीन < p-टॉलूडील

(b) $C_6H_5NH_2 < C_6H_5NHCH_3 < C_6H_5CH_2NH_2$

(iv) $(C_2H_5)_3N > (C_2H_5)_2NH > C_2H_5NH_2 > NH_3$

(v) $(CH_3)_2NH < C_2H_5NH_2 < C_2H_5OH$

(vi) $C_6H_5NH_2 < (C_2H_5)_2NH < C_2H_5NH_2$

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